



ELITE™ AT 6112S

Enhanced Polyethylene Resin

Overview

ELITE™ AT 6112S Enhanced Polyethylene Resin is an ethylene-hexene copolymer based on ELITE™ AT technology from Dow. This resin is designed for cast stretch machine wrap films in pallet wrap application and offers an optimized balance of extensibility, impact strength, and puncture resistance.

Main Characteristics:

- Excellent extensibility
- High impact and exceptional puncture resistance
- Excellent processibility with low back pressure and power consumption.

Complies with:

- U.S. FDA FCN 1539

Consult the regulations for complete details

Additive

- Antiblock: No
- Slip: No
- Processing Aid: No

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density	0.910 g/cm ³	0.910 g/cm ³	ASTM D792
Base Density ¹	0.912 g/cm ³	0.912 g/cm ³	Dow Method
Melt Index (190°C/2.16 kg)	3.8 g/10 min	3.8 g/10 min	ASTM D1238
Films	Nominal Value (English)	Nominal Value (SI)	Test Method
Film Thickness - Tested	1.0 mil	25 µm	
Film Puncture Energy (1.0 mil (25 µm))	56.8 in·lb	6.42 J	Dow Method
Film Puncture Force (1.0 mil (25 µm))	8.30 lbf	36.9 N	Dow Method
Film Puncture Resistance (1.0 mil (25 µm))	371 ft·lb/in ³	30.7 J/cm ³	Dow Method
Tensile Strength			ASTM D882
MD : Yield, 1.0 mil (25 µm)	982 psi	6.77 MPa	
TD : Yield, 1.0 mil (25 µm)	930 psi	6.41 MPa	
MD : Break, 1.0 mil (25 µm)	7370 psi	50.8 MPa	
TD : Break, 1.0 mil (25 µm)	6170 psi	42.5 MPa	
Tensile Elongation			ASTM D882
MD : Break, 1.0 mil (25 µm)	550 %	550 %	
TD : Break, 1.0 mil (25 µm)	610 %	610 %	
Dart Drop Impact (1.0 mil (25 µm))	500 g	500 g	ASTM D1709A
Elmendorf Tear Strength ²			ASTM D1922
MD : 1.0 mil (25 µm)	360 g	360 g	
TD : 1.0 mil (25 µm)	520 g	520 g	
Ultimate Stretch - On-Pallet Testing ³ (1.0 mil (25.4 µm))	380 %	380 %	Dow Method
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Melting Temperature (DSC)	228 °F	109 °C	Dow Method
Optical	Nominal Value (English)	Nominal Value (SI)	Test Method
Gloss (45°, 1.00 mil (25.4 µm))	93	93	ASTM D2457
Haze (1.00 mil (25.4 µm))	1.06 %	1.06 %	ASTM D1003

Extrusion Notes

Fabrication Conditions For Cast Film:

- EGAN/Davis-Standard 5 layer cast line
- Melt Temperature: 520°F (271°C)
- Chill Roll (primary/secondary) Temperature: 70°F (21°C)
- Line Speed: 600 fpm (183m/min)
- Output: 400 lb/hr
- Die Width: 36in. (915 mm)
- Die Gap: 20mil (0.5mm)
- Air Gap: 3in. (7.6mm)

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ Base density is estimated using the assumption that every 1000 ppm of antiblock in the finished product raises the density of the polymer by 0.0006 g/cm³. Base density is the estimated density of the polymer if it did not contain any antiblock.

² Method B

³ Determined by Highlight Industries, Inc.

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